

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

CSA1008 - Software Engineering

Assignment Title: CO4-AT1-Test Case Design Assignment

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Case Study Title: Flood Early Warning System

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1. Introduction

The **Flood Early Warning System** is a software solution designed to monitor water levels in rivers, reservoirs, and urban drainage channels to provide timely alerts to authorities and citizens. The system continuously tracks water levels, identifies critical flood thresholds, generates optimized evacuation and response routes for rescue teams, and records historical flood data. This helps minimize loss of life, prevent severe infrastructure damage, reduce emergency response delays, and lower operational mitigation costs.

The purpose of testing is to verify the correctness, reliability, security, usability, and efficiency of the proposed system under normal, invalid, boundary, and exceptional conditions.

2. Functional Requirements

ID	Functional Requirement
FR01	The system shall allow authorized users (operators/administrators) to log in.
FR02	The administrator shall be able to register new water level monitoring sensors/stations.
FR03	The system shall monitor the water level/river depth continuously.
FR04	The system shall identify flood risk levels requiring public warnings.
FR05	The system shall generate automated alert broadcasts for critically high water levels.
FR06	The system shall generate optimized evacuation and response routes.
FR07	The system shall allow emergency workers to update flood mitigation and status reports.
FR08	The system shall record completed evacuation and mitigation activities.
FR09	The administrator shall be able to view historical flood and sensor log data.

3. Non-Functional Requirements

ID	Requirement
NFR01	The system should provide fast response to real-time sensor updates and user requests.

ID	Requirement
NFR02	The system should allow only authorized users to access protected system functions.
NFR03	The system should operate reliably with high availability during extreme weather events.
NFR04	The system should be easy for operators, local authorities, and field personnel to use.
NFR05	The system should display accurate real-time water level and status information.

4. Test Scenarios

Scenario ID	Test Scenario
TS01	Verify login using valid credentials.
TS02	Verify login using invalid credentials.
TS03	Verify registration of a new flood monitoring sensor.
TS04	Verify monitoring of water level readings.
TS05	Verify identification of flood warning thresholds.
TS06	Verify critical flood alert generation and broadcast.
TS07	Verify invalid water level sensor input handling.
TS08	Verify emergency evacuation route generation.
TS09	Verify rescue/mitigation status update.
TS10	Verify completion of flood response activity.
TS11	Verify flood activity and sensor historical logs.
TS12	Verify unauthorized access prevention.

5. Detailed Test Cases

Assumptions & Threshold Rules:

- Water Level 0 - 79% (Safe Level): Flood alert not required.
- Water Level 80 - 94% (Warning Level): Flood warning required.
- Water Level 95 - 100% (Critical Flood Level): High-priority emergency alert required.

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual	Status
TC01	Valid Login	Enter username & password → Click Login	Valid credentials	User is successfully logged in	Not Executed	Not Executed
TC02	Invalid Login	Enter incorrect password → Click Login	Valid user + wrong password	Login rejected; error displayed	Not Executed	Not Executed

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual	Status
TC03	Register Sensor	Enter sensor ID & location → Save	SENSOR101, River Basin Alpha	New sensor registered successfully	Not Executed	Not Executed
TC04	Normal Water Level	Enter/update water level	50%	Water level accepted; status safe	Not Executed	Not Executed
TC05	Warning Threshold	Update water level	80%	Identified as Flood Warning status	Not Executed	Not Executed
TC06	Critical Level	Update water level	95%	Critical flood alert generated	Not Executed	Not Executed
TC07	Invalid Level	Enter invalid level	110%	Rejects value; validation error	Not Executed	Not Executed
TC08	Evacuation Route	Select flood zones → Generate Route	SENSOR101, SENSOR102	Evacuation/response route generated	Not Executed	Not Executed
TC09	Update Status	Worker opens route → Updates status	In Progress	Status changes to 'In Progress'	Not Executed	Not Executed
TC10	Complete Mitigation	Select completed area → Complete	SENSOR101 Zone	Recorded as completed	Not Executed	Not Executed
TC11	View History	Open flood history → Select date/ID	SENSOR101	Historical flood data displayed	Not Executed	Not Executed
TC12	Unauthorized Access	Normal user attempts admin function	Normal account	Access denied	Not Executed	Not Executed

6. Test Design Techniques

6.1 Equivalence Partitioning

Equivalence Partitioning divides input values into valid and invalid groups. For the water level sensor inputs:

Partition	Range	Example	Expected Result
Invalid	Less than 0%	-10%	Reject
Valid	0 - 79%	50%	Safe Level (No Alert)
Valid	80 - 94%	85%	Warning Level (Issue Warning)
Valid	95 - 100%	98%	Critical Flood Level (Emergency Alert)
Invalid	Greater than 100%	110%	Reject

7. Boundary Value Analysis

Boundary Value Analysis checks system behavior at and around critical boundary limits.

Test Value	Expected Result
-1%	Reject
0%	Accept (Safe)
79%	Safe Level (No Alert)
80%	Warning Level (Issue Warning)
81%	Warning Level (Issue Warning)
94%	Warning Level (Issue Warning)
95%	Critical Level (Emergency Alert)
100%	Critical Level (Emergency Alert)
101%	Reject

8. Decision Table Testing

The system determines emergency procedures based on detected water level ranges:

Condition / Action	Rule 1	Rule 2	Rule 3
Water Level	< 80%	80 - 94%	95 - 100%
Warning Required	No	Yes	Yes
Critical Alert	No	No	Yes
Priority	Normal	Medium	High

Examples:

- **50%:** Safe level (No Alert)
- **85%:** Warning Alert Issued
- **97%:** High-priority Critical Flood Alert + Emergency Evacuation Triggered

9. State Transition Testing

The monitored flood zone shifts through distinct operational states based on water level conditions:

Normal → Flood Warning → Critical Flood → Response In Progress → Flood Subsided → Normal

Current State	Event	Next State
Normal	Water level reaches 80%	Flood Warning
Flood Warning	Water level reaches 95%	Critical Flood
Warning / Critical	Emergency team dispatched	Response In Progress
Response In Progress	Evacuation completed	Flood Subsided

Current State	Event	Next State
Flood Subsided	Water level drops below 80%	Normal

10. Requirement Traceability Matrix

Requirement	Test Case IDs
FR01 - User Login	TC01, TC02
FR02 - Sensor Registration	TC03
FR03 - Water Level Monitoring	TC04, TC05, TC06, TC07
FR04 - Warning Threshold Identification	TC05
FR05 - Critical Alert Generation	TC06
FR06 - Route Generation	TC08
FR07 - Response Status Update	TC09
FR08 - Response Record	TC10
FR09 - History & Log Viewer	TC11
NFR02 - System Security	TC12

11. Test Coverage

Requirement Coverage

Total Functional Requirements = 9 | Requirements Covered = 9

Requirement Coverage = $(9 / 9) \times 100 = 100\%$

Scenario Coverage

Total Test Scenarios = 12 | Scenarios Covered = 12

Scenario Coverage = $(12 / 12) \times 100 = 100\%$

Condition Coverage

Test Condition	Coverage
Normal Condition	Covered
Invalid Condition	Covered
Boundary Condition	Covered
Exceptional Condition	Covered
Security Condition	Covered

Test Condition	Coverage
State Transition	Covered
Business Rules	Covered

12. Conclusion

The **Flood Early Warning System** has been thoroughly analyzed, and a structured test case design has been prepared to verify its functionality, reliability, security, and usability under critical emergency conditions.

Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing techniques were applied to optimize test coverage. The Requirement Traceability Matrix validates complete mapping between requirements and test cases. The test suite guarantees that the system can reliably track water levels, broadcast emergency alerts, route evacuation teams, and support critical climate action decision-making.